

Windows of opportunity: white-tailed deer and the dynamics of northern hardwood forests of the northeastern US

Richard W. Sage Jr. ^{†1}, William F. Porter^{1,*} & H. Brian Underwood²

¹Adirondack Ecological Center, State University of New York College of Environmental Science and Forestry, Syracuse, 6312 Route 28 N, Newcomb, New York, USA 12852, e-mail: wporter@esf.edu

²United States Geological Survey Biological Resources Division, Patuxent Wildlife Research Center, State University of New York College of Environmental Science and Forestry, Syracuse, New York, USA

Abstract

Herbivory, lighting regimes, and site conditions are among the most important determinants of forest regeneration success, but these are affected by a host of other factors such as weather, predation, human exploitation, pathogens, wind and fire. We draw together >50 years of research on the Huntington Wildlife Forest in the central Adirondack Mountains of New York to explore regeneration of northern hardwoods. A series of studies each of which focused on a single factor failed to identify the cause of regeneration failure. However, integration of these studies led to broader understanding of the process of forest stand development and identified at least three interacting factors: lighting regime, competing vegetation and selective browsing by white-tailed deer (*Odocoileus virginianus*). The diverse 100–200 year-old hardwood stands present today probably reflect regeneration during periods of low deer density (<2.0 deer/km²) and significant forest disturbance. If this hypothesis is correct, forest managers can mimic these “natural windows of opportunity” through manipulation of a few sensitive variables in the system. Further, these manipulations can be conducted on a relatively small geographic scale. Control of deer densities on a scale of 500 ha and understory American beech (*Fagus grandifolia*) on a scale of <100 ha in conjunction with an even-aged regeneration system consistently resulted in successful establishment of desirable hardwood regeneration.

Key words: American beech, *Fagus grandifolia*, northern hardwoods, *Odocoileus virginianus* regeneration, shelterwood, white-tailed deer.

Introduction

A central question in silviculture throughout the northern hardwood forests of eastern North America is how to regenerate a forest that is similar to the diverse hardwood communities that we have today. Our goal was to regenerate stands to a high diversity of desirable plant species at high enough stem densities to ensure good development. This goal is consistent with both our timber management objective of growing high quality sawlogs and veneer logs; and our wildlife management objective to increase representation of both commercial and non-commercial fruit-bearing trees, shrubs and herbs in new stands.

Our interest in the ecology of regeneration was heightened by our repeated failure to achieve satisfactory results in our early attempts. Application of traditional silvicultural techniques designed to perpetuate an uneven-aged forest by employing the selection system (Society of American Foresters 1971) resulted in a near monoculture of shade-tolerant, slow-growing, low commercial value American beech that competitively excluded more desirable hardwood species. The extremely high shade tolerance of beech, coupled with its unpalatability to deer and aggressive root-suckering ability, favors this species above all others in closed

*corresponding author

canopy forests with an abundance of deer. The arrival of beech bark disease (*Nectria coccinea*, and *Cryptococcus fagisuga*) in the early 1960s exacerbated the rate at which beech was engulfing the understories of our hardwood stands. Mature beech stems produced an abundance of new root suckers prior to succumbing to this disease (Sage 1996). The increasing dominance of poor quality, low value beech as a replacement for sugar maple (*Acer saccharum*), yellow birch (*Betula allegheniensis*), black cherry (*Prunus serotina*), white ash (*Fraxinus americana*), and other high value species clearly called for an alternative management approach.

Identifying that alternative approach required more than 30 years of study, and exploration of lighting regimes, dynamics of deer populations, and competition among plant species. Our objective here is to summarize the principal messages from these various studies and describe the practical forest management program that arose from a synthesis of the findings.

Study area

The studies were conducted on the Huntington Wildlife Forest (HWF), (lat. 43-59-05, long. 074-14-06). This property is a 6,000 ha field research station of the State University of New York, College of Environmental Science and Forestry. Ecological research on issues related to forestry, wildlife, soils and hydrology have been ongoing since 1932.

The HWF is located in the geographic center of the Adirondack Park, a 24,281 km² portion of northern New York State. With the exception of the widely scattered hamlets, and the thousands of small ponds and lakes, the region is entirely forested. A near equal proportion of privately owned forest land and public forest land are interspersed throughout the Park. Private lands support a forest-product based economy that is active, but shrinking. Public lands are protected as "forever wild" and contribute primarily to an expanding outdoor recreation-based tourism economy.

Flora and fauna are transitional between the deciduous and taiga biomes (Dice 1952; Allee et al. 1949). The growing season is 90–120 days. The mean January temperature is –8.1 °C, and the mean July temperature is 18.2 °C. Mean annual precipitation is 107 cm distributed evenly throughout the year. The average annual total snowfall is 308 cm. Topography is mountainous with elevations ranging from 457 m to 853 m. The combination of elevation, snowfall and cold temperatures generally results in a continuous snowpack of 38–152 cm from late December through the end of March. Soils are of glacial origin and are generally shallow, rocky and low to moderate fertility.

The indigenous hardwood tree species of commercial importance include; sugar maple, yellow birch,

American beech, red maple (*Acer rubrum*), black cherry, white ash, basswood (*Tilia americana*) and paper birch (*Betula papyrifera*). The more common conifer species include red spruce (*Picea rubens*), balsam fir (*Abies balsamifera*), Eastern hemlock (*Tsuga canadensis*), and white pine (*Pinus strobus*). The three most common forest communities occurring on the study area include Type 25, Sugar maple – Beech – Yellow birch; Type 24, Hemlock – Yellow birch; and Type 33, Red spruce – Balsam fir (Society of American Foresters 1980).

The white-tailed deer is the dominant large herbivore, although moose (*Alces alces*) have been invading the system during the past decade from nearby populations in Canada and New England. Currently however, their numbers are minimal. The coyote (*Canis latrans*) invaded the region in the 1950s, dramatically expanding its presence through the 1970s and currently represents the dominant year-round predator on white-tailed deer. The black bear (*Ursus americanus*) is also an important predator, especially during the spring period (Mathews & Porter 1988). Wolves (*Canis lupus lycaon*) and mountain lions (*Felis concolor*) were extirpated in the mid 19th century (Wilson et al. 2000).

Hunting represents a significant contribution to the fall economy of the region. Both New York State residents and hunters from across the northeast come to the region in pursuit of both white-tailed deer and black bear. A significant portion of the land owned by large timber companies in the region is leased for deer and bear hunting. Deer hunting is regulated by the New York State Department of Environmental Conservation and has been restricted to "bucks-only" hunting in the Adirondack region since 1971. A limited harvest of antlerless deer occurs each year during archery and muzzleloader seasons. We operated experimental deer hunting programs on HWF under special permit authority (buck hunting in conjunction with antlerless deer permits), or in concert with these primitive weapons seasons (either sex hunting).

Exploring lighting regime

During 1943 through 1954, four mature northern hardwood stands (80–200 ha) on HWF were logged to remove 100%, 75, 50 and 25 of the merchantable volume (mean BA/ha = 31.0 m²). These harvest rates should not be equated with basal area (BA) removal. For example, the BA after cutting at the 100% volume removal site was still 11.9 m²/ha, comprised largely of poor quality hardwoods and advance beech regeneration.

Tierson (1967) reported on the 75% removal site, and showed that beech increased in density following logging to the extent that 90% of the 2.5–10.2 cm

DBH class was comprised of this species. He concluded that beech sapling and pole-sized trees so dominated the understory that other more desirable species had not formed a silviculturally significant part of the vegetation in these size classes. Similar responses were documented at the other three study sites (Krull 1963, Huntington Forest unpublished data). In all cases, the presence of advance beech regeneration prevented the establishment and development of adequate numbers of desirable hardwood species.

In 1957–58 a series of 0.81 ha blocks, representing three replications of five overstory cutting levels were installed at HWF (Barrett et al. 1962; Farnsworth & Barrett 1966; Farnsworth & Richards 1971). In these trials, BA was used to describe the residual density of the overstory following harvest. Cutting levels were uncut, 0, 7.1, 11.9 and 16.7 m²/ha residual BA. In addition to the overstory cutting, one half of each plot was treated with an herbicide (2,4,5-T in oil) as a basal spray to control all understory vegetation 1 m high to 11.4 cm DBH. Four deer exclosures were constructed in each block, two on the herbicide treated portion of the block, and two on the untreated portion.

These early experiments focused primarily on defining the appropriate lighting regime necessary to stimulate a hardwood regeneration response. These trials ultimately led to the selection of 7.1–14.3 m²/ha residual BA as the overstory density which resulted in a well-stocked, diverse tree species response. However, the ability to evaluate the regeneration response to the various overstory cutting treatments was only made possible by incorporation of herbicide treatments and deer exclosures into their experimental design due to complete domination by beech on the other portions of the study blocks. Failure to address these two factors produced inconsistent regeneration responses regardless of the cutting intensity.

Controlling competing vegetation

At the ground level, regeneration was affected by competing vegetation, primarily American beech. Studies by Tierson et al. (1967) showed that understory control of beech prior to cutting was necessary to assure adequate establishment and growth of desirable hardwood species. Behrend & Patric (1969) examined this understory from the perspective of food availability to deer and documented the negative effects of shade cast by the beech understory on the establishment of deer browse.

In the 1980s, studies showed that the aggressive competitive ability of beech was a product of natural history and selective browsing. The extreme shade tolerance of beech and its ability to propagate by root-suckers from a parent tree positioned in the overstory

contributed to the dominance of beech at the ground level (Jones & Raynal 1988; Jones et al. 1989). As a browse, beech is generally avoided by white-tailed deer in the Adirondacks (Frank 1940). This combination promoted densities of 2471–4448 regenerating beech stems/ha. Dominance of beech in the understory of mature hardwood stands resulted in near-exclusion of most other hardwood species (Tierson 1967; Sage 1987).

Trials to control beech focused first on small plots and later at operational scales. On small research plots, beech saplings were treated individually using a variety of chemicals either by basal spray or direct injection in spaced-axe cuts. On operational-scale cutting units (>25 ha) these techniques were inefficient and costly. To reduce cost, application was adapted for broadcast treatment, initially with a back-pack mist-blower and eventually using a large mist-blower unit mounted on the rear of a rubber-tired logging skidder (Tierson 1969a; Sage 1987). The skidder unit could treat up to 2.4 ha/hour and achieve effective control up to 9.1 m in height. Glyphosate at a rate of 2.7 L in 66 L of water/ha consistently produced a kill rate of >85% on beech trees up to 10.2 cm DBH (Sage 1983). This approach reduced costs to \$86 to 161/ha. Typically the beech understory was treated prior to cutting trees in the main canopy. This approach relied on the beech sapling layer to intercept the chemical and prevent it from reaching taller, merchantable trees (Sage 1987).

Understanding deer populations

Coincident with the research on the influence of lighting and competition from beech was a series of studies that sought to understand the population ecology of deer, and methods for controlling deer densities. Densities of deer on HWF were estimated using a variety of techniques including deer drive counts, track counts and visual observation rate surveys along a 40.2 km gravel road system. The earliest estimates placed deer densities of the HWF at 10/km² in 1946–47 (Steinhoff 1947). This estimate was based on three “drive counts” made during the fall, pre-hunting season.

Initial attempts at deer density control at HWF at an operational scale involved use of electric fencing. These efforts proved extremely costly and time consuming and provided inadequate control when large management units were involved (Tierson 1969b).

In the fall of 1966 the first of two experimental deer hunts on HWF was initiated. Pre-hunt densities were estimated at 10.4/km². Impacts of deer on vegetation were severe, with some sites failing to develop any tree regeneration for up to 15 years post cutting. Uncertain of just what deer density would be compatible with timber management objectives, a goal to reduce

deer numbers by one half (to 5.2/km²) was established. The pre-hunt deer population on the hunting unit was estimated from "drive counts" conducted on 10% of the hunting area. This number was then used to develop a fall, pre-hunt population estimate based upon deer distribution across the entire 21 km² management area.

Hunting was conducted using modern firearms under special permit. Careful management of hunter numbers and deer harvest resulted in achieving the desired population reduction at the end of the second hunting season. Three subsequent years of hunting focused on removing the annual increment of animals to the population and sustaining the desired deer density. An aggressive harvest of female deer under a special permit system resulted in a near equal harvest of males and females over the course of the 5-year hunting program (Behrend et al. 1970; Underwood 1986).

Following reduction in deer densities the vegetation response was dramatic. Areas that had been in raspberry (*Rubus* spp.) for years immediately began to return to forest cover. Stands of heavily browsed maple responded rapidly and had outgrown deer influence in <5 years. Highly palatable tree species like yellow birch, that had long been absent, regenerated in abundance after deer densities had been reduced. Over the next 10 years, studies showed that lowered deer densities would consistently promote successful establishment of diverse, well-stocked, hardwood regeneration on units > 30 ha in size (Behrend et al. 1970; Kelty & Nyland 1981, 1983).

Unfortunately, the opportunity to identify a clear threshold density allowing regeneration was eliminated by a stochastic weather pattern. During 1969 through 1971, 3 successive severe winters caused substantial mortality and the deer population estimate fell from 4.6 deer/km² to <1.9 deer/km². The deer population expanded through the 1970s, rising to 5.8 deer/km² in 1981.

Drive counts in combination with visual observation rates of deer along forest roads were used to estimate the population during the post 1971 period (Sage et al. 1983).

A second public hunt was conducted during 1978 through 1983 to maintain current deer densities and prohibit a rapid increase in numbers. The use of primitive weapons hunters (archery and muzzleloader), without the requirement for special permits to harvest female deer proved equally effective in preventing rapid growth in deer numbers beyond the "targeted" population density (5.2 deer/km²). These hunts also showed that significant income (\$5–12/ha) could be generated through controlled access public hunting as part to a well-managed deer density control program (Sage & Porter 1987).

Studies of an adjacent unhunted population on HWF showed that deer densities fluctuated widely primarily in response to winter severity. The principal factor driving this fluctuation was the length of time deer were confined by deep snow to winter range. Populations grew only during periods of time when winters were milder than average. During average or severe winters, populations remained constant or declined (Underwood 1990). Over a 40-year time span, mild winters occurred with a frequency of about 3 in 10. Development of desirable forest regeneration on sites where overstory tree densities were reduced and understory beech eliminated occurred consistently across HWF when local deer densities were <5.2 deer/km²; regardless of whether population reduction resulted from direct intervention (i.e., hunting) or severe winter weather (Kelty & Nyland 1981; Underwood 1986).

The greatest impacts of deer on northern hardwood regeneration proved to be associated with browsing during the latter half of summer and throughout the fall when deer were on their summer-fall range. Individual adult deer in northern forested environments utilize specific summer (spring/summer/fall) and winter ranges throughout their lives. These seasonal home ranges were shown to be typically separated by several miles. Fidelity of individual deer to their established home ranges was high (Tierson et al. 1985; Mathews 1989).

As yearlings, female offspring of adult does established adjacent and overlapping summer home ranges with their parent and migrated to the same wintering area used by the parent doe. Once established this seasonal movement behavior was repeated each year throughout the remainder of an animal's life. Progressive generations of young females establishing overlapping ranges with their mothers generated familial associations of female deer occupying distinct parcels of geography on the landscape. While overlap of home ranges of individual females within a matriarchal group was considerable, overlap between adjacent female groups is minimal. Males associated with these matriarchal groups were either pre-dispersal offspring, or post-dispersal adults which established a permanent home range in the area (Tierson et al. 1985; Mathews & Porter 1991; Aycrigg & Porter 1997).

The potential importance of this seasonal ranging behavior became evident in a removal experiment conducted during the 1990s. A set of seven matriarchal groups of deer was radio-collared and 14 of 17 deer in one group were removed. This removal created an area of 150 ha of very low deer density (Porter et al. 1991; McNulty et al. 1997). The hypothesis, prompted by the high fidelity exhibited by adult females to their established home ranges, was that deer from other matriarchal units would not move into this area of low density. Five years of subsequent monitoring showed that deer did not move and the reduced density area persisted.

The distribution and relative “success” of these matriarchal groups could explain widely varying local impacts on forest vegetation across a broader landscape.

Synthesis

The insight gained through years of research into the critical factors leading to successful regeneration of the northern hardwood forest led us away from a traditional uneven-aged management system using single-tree selection to an even-aged system of management using the shelterwood regeneration method. The uneven-aged approach was not wrong, it just wasn't right in the context of the complex set of factors influencing regeneration in the central Adirondacks. Our regeneration problems were linked to canopy closure, competition from dense beech understories, and too many deer. How we managed one dimension of this three-faceted system made little difference until we dealt effectively with all dimensions.

This complexity led to the selection of an even-aged management system. Control of beech understories using chemical treatments at each entry into a multiple-aged stand without damaging established age classes would be nearly impossible and very expensive. The added advantage of the even-aged system was that heavier cutting encouraged shade intolerant and mid-tolerant plant species. This was important to both our timber and wildlife management objectives (Nyland 1987; Porter 1987). Lastly, operating large equipment in the rugged terrain typical of our region required room to maneuver. Heavier cutting created larger corridors and minimized felling and skidding damage to residual trees (Nyland & Gabriel 1971). It also reduced the number of entries onto a site in any single rotation minimizing impacts to soils and drainage. In addition, this reduced frequency of entry is more consistent with maintaining important wildland values than traditional short-term cutting cycles (Brocke et al. 1991).

Including deer population dynamics and behavior into our timber management regime proved crucial. On large scales (> 10,000 ha), regulated harvest of deer is not a management option because the bucks-only hunting restriction precludes effective population control. However, winter conditions throughout much of the Adirondack region produce fluctuation (Underwood 1990; Nesslage & Porter 2001). Periods when deer populations are low constitute windows of opportunity for managers.

On smaller scales, the ranging behavior of deer allows good potential for management at the stand or multi-stand level. Hunting during the regular fall big-game season allows targeting those deer that are im-

pacting the forest vegetation on the management unit of concern. Harvesting antlerless deer in this area will reduce the number of deer on the unit of interest because each matriarchal group occupies a relatively distinct portion of the landscape. The high level of fidelity to home range means that the low density area can persist even in the face of higher deer densities on the surrounding landscape. If the low density persists for at least five years, then regeneration can grow out of the reach of deer when appropriate lighting conditions are provided (Behrend et al. 1970; Kely & Nyland 1981).

Localized deer density control can create a window of opportunity on a planned schedule. The availability of antlerless deer permits issued directly to a private landowner with an approved forest management program under the New York State Department of Environmental Conservation's "Deer Management Assistance Program" should facilitate effective and efficient localized deer density control. Harvesting, site preparation treatments (if necessary), road construction, and deer density control operations must be well coordinated. Site preparation and deer density control must be in place before cutting begins to ensure establishment and development of the most vulnerable species. A well-conceived and implemented program however, will permit the forest manager to create a window of opportunity and take control of another critical element contributing to a successful regeneration effort.

In summary, the basic operational steps in the regeneration system for northern hardwoods in the central Adirondack region of New York which eventually evolved from the research conducted at HWF can be characterized as follows:

1. Local deer density control is required if deer populations exceed 5.2/km² (Behrend & Patric 1969; Behrend et al. 1970).
2. Beech understory control using broadcast chemical treatment is necessary prior to cutting if the number of beech stems 2 m–10.2 cm DBH exceeds 1235/ha (Tierson 1967; Sage 1987).
3. Lighting regimes and seed source are controlled using the shelterwood regeneration method. The entire mature stand is removed in two cuttings. The initial cutting (shelterwood establishment cut) reduces residual BA by 50–65% (9.5–14.3 m²/ha). The second cutting removes the remainder of the original stand (removal cut), and is made after the newly regenerated even-aged forest has developed beyond the reach of deer (typically 5–10 years after the shelterwood cut) (Kely & Nyland 1981).

Stands regenerated using this system consistently include 8–10 commercial tree species with sugar maple and yellow birch comprising up to 90% of the stocking. In addition, 4–6 non-commercial tree species are represented as well as a variety of fruit-bearing shrubs during the first 10–15 years post cutting. At age

5–7 the new even-aged stand typically supports >80,000 stems/ha. Stem counts at 20–5 years of age range from 7600–8900 stems/ha with shade intolerant species >18 cm DBH and mid-tolerant species averaging 9–12 cm DBH (Nyland et al. 2000).

Integrations and conclusions

While many of the findings from the studies at HWF have specific value to the Adirondacks, the lessons learned have much broader implications. First, the combined power of the studies shows that stand establishment and development in the northern forest is driven by a complex interaction of multiple factors. We must be cautious about becoming fixated on a single cause for regeneration failure. The studies conducted on HWF were each initiated with the motive of testing a single factor: first lighting regime, then competing vegetation, and then browsing by deer. Patten et al. (2002) illustrates and amplifies this message with his analogy to the dangers of flying an airplane by focusing on a single instrument. Browsing by ungulates is often construed as the dominant factor in regeneration failure, but browsing pressure is not uniform on a landscape scale (Didier & Porter 2002). Only when each factor is viewed in the context of a system of factors, can we best discern its role.

Second, regeneration of northern hardwoods forests may not be a continuous process, but more episodic. We were schooled to perceive systems in equilibrium and to view ecological processes as on-going at all times. The gap-phase replacement model, and the single-tree selection regeneration method are perhaps extensions of such thinking, but may not be as applicable to these forests as we once thought. This is particularly

apparent when we include complicating factors such as beech bark disease, and stochastic events like mild or severe winters.

The influences of these factors are less direct, but produce effects that ultimately translate into fluctuation in forest regeneration responses.

As a way of capturing the complexity of multi-factor interactions, we present a conceptual model for predicting the stocking of stems of desirable hardwoods in the northern hardwood forest (Figure 1). We are currently parameterizing a more sophisticated model for research hypothesis generation and testing. The basic model presented here has three primary stocks: deer abundance, competing vegetation, forest regeneration and several intermediate state variables that characterize the system at any given point in time. It summarizes years of research findings and emulates the regenerative processes of the northern hardwood forest after canopy disturbance, control of competing vegetation and deer browsing.

White-tailed deer are key regulators in the forest renewal process primarily through their browsing habits, which differentially alter the relative competitive abilities of young trees. Years of selective browsing by deer can render an understory to a near monoculture of highly resistant or unpalatable species. It is this understory of advanced regeneration that will become the forest of tomorrow. In our conceptual model, understory species composition is modeled as a function of deer abundance (Figure 1A).

Deer populations in this environment have to contend with the vagaries of harsh winter weather, and predation by both coyotes and humans. Deer deal with winter by establishing disjunct home ranges, from which they migrate between seasons. If migration occurs prior to 1 January, the population is expected to decline as a

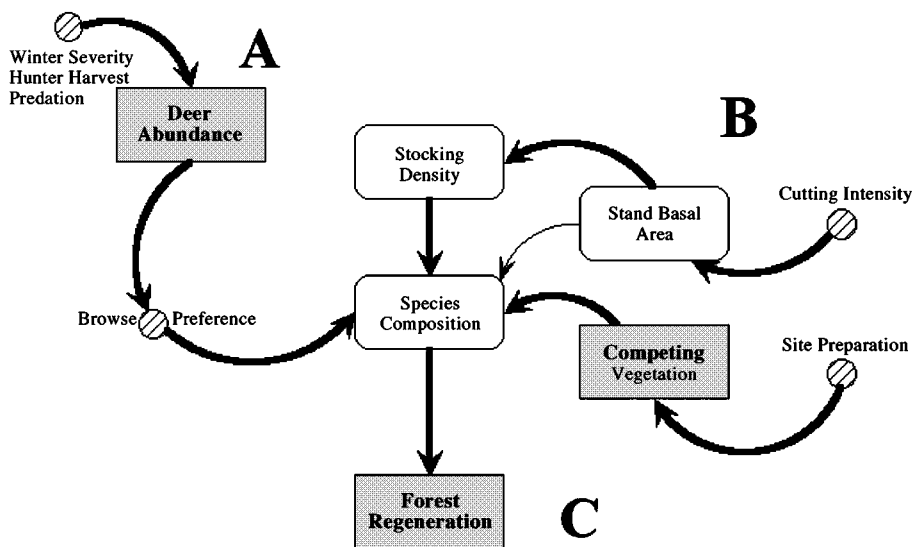


Figure 1. Conceptual model of the process of northern hardwood regeneration and the principal factors affecting its development. The three primary stocks are shown as rectangles, the intermediate state variables are shown as rounded rectangles, and the control variables are shown as circles. Letters A (deer abundance), B (index to light based on basal area of the stand) and C (index of competing vegetation) designate the component processes leading to regeneration.

result of a negative rate of change. Hunter harvest and predation by coyotes occurs every year, but rarely do either account for more than 25% of the total annual mortality. Because winter weather drives abundance, deer population rate of change can be modeled as a numerical response to winter severity, which obviates the need to model birth and death rates explicitly. Likewise, changes in abundance due to predation and hunter harvest can be incorporated as growth-rate multipliers.

Forest renewal is accomplished though the conversion of biomass in the form of relatively few, large trees to an abundance of younger ones. Large trees process most of the available light and nutrients at locations that are not susceptible to deer herbivory. Regenerants, however, remain within the reach of deer for perhaps a decade, which underscores the role of herbivory in the composition and structure of future forests. Light within a forest stand is determined largely by the degree to which the forest canopy is developed. Canopy development is a function of tree size, which we index by the stand's basal area (Figure 1B). Basal area of overstory or canopy trees dictates the abundance of advanced regeneration in the understory.

Logging alters the basal area as trees are removed and canopy gaps are created, thus changing the relative capacity of the stand to support advance regeneration. A close relationship has been demonstrated between the stocking density in the understory and stand basal area. This relationship is modeled from an equation predicting the number of seedlings and saplings that might occupy the canopy in the future. Water is usually in ample supply in the northeastern United States; therefore, competition for sunlight is probably the single most important limitation on plant growth. Basal area in our conceptual model functions as a measure of the intensity of competition for this resource.

Depending upon the levels of herbivory in the recent past (1–3 decades), the forest understory may be dominated by species that are not desirable from an economic, aesthetic, or ecological perspective (Figure 1C). Given the opportunity to grow, these species will dominate the canopy of future forests unless, through intervention, the species composition and other factors are modified appropriately. Establishment of satisfactory forest regeneration without dealing with previously existing undesirable vegetation, even with deer density control, will produce a poor result. Because current methods of understory control eliminate nearly all understory vegetation, the abundance of a new regeneration cohort is a function of post-treatment germination processes and time since cutting. Combined with intensive cutting, this system produces a regenerating stand with a diverse hardwood species composition and abundant stocking. These stands must be protected from browsing until the regenerants grow beyond the reach of deer, which can take from 5–10 years post-cutting.

Although complex, the system we describe can be managed. The integration of the findings from the various studies tells us why we failed to regenerate a diverse stocking of hardwoods. Development of comprehensive, ecology-based models of the mechanisms controlling regeneration of forest stands helps us understand how we failed, and more importantly, how to manipulate the system to promote success. The experience of continually adapting our management shows us that we can manipulate the system to achieve desired outcomes.

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